

Ayush Bajaj

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EDUCATION

- **VIT University, Vellore** 2023 – 2027
Bachelor of Technology, Computer Science CGPA: 8.76

SKILLS

- **Programming Languages:** Python, Java, C, HTML, CSS
- **Cloud Computing:** AWS (EC2, IAM, Amazon Polly), Cloud Deployment, Scalable Architecture
- **Web Technologies:** React.js, Node.js, Express.js, REST APIs, Flask
- **Machine Learning:** Classification Models, Feature Engineering, Model Evaluation, Transformer Models (FLAN-T5)
- **Databases:** MySQL, MongoDB, Neo4j
- **Core Computer Science:** Data Structures and Algorithms, OOPs, DBMS
- **Tools:** Git, GitHub, GitHub Actions, VS Code, FFmpeg, Boto3, Docker
- **Soft Skills:** Problem Solving, Critical Thinking, Time Management, Adaptability

EXPERIENCE

- **KuppiSmart Solutions** May 2025 – Jul 2025
Software Developer Intern
 - Built 10+ responsive web pages and a reusable component library using React.js and Tailwind CSS, reducing layout shift by an estimated 30% through optimized render paths and lazy loading.
 - Integrated the Razorpay payment gateway through secure REST APIs, supporting an estimated 500+ transactions across 2-week agile sprints, and shipped 3 iterative feature releases.
 - Automated build, test, and deploy pipeline using GitHub Actions, cutting manual release time by an estimated 40% and enforcing consistent code-style guidelines across a 5-member engineering team.

PROJECTS

- **Microservice Drift Tracker (MDT)**
Distributed Systems Developer | Python (FastAPI), Neo4j, Docker, React
 - Built an automated architectural governance engine (HMDA – Hierarchical Microservice Drift Analysis) that statically analyzes Git commits across microservices using AST parsing in 8 languages and Neo4j graph traversal to flag breaking contract changes and 10 distributed architectural anti-patterns before deployment.
 - Containerized the full platform (FastAPI, Neo4j, ChromaDB, React) with Docker Compose, computing a normalized 0–100 risk score and shipping Git pre-commit/pre-push hooks with GitHub webhook integration, validated by 24/24 automated tests.
- **AI-Based PPT Explainer & Video Generator**
Cloud & NLP Developer | AWS, FLAN-T5, Amazon Polly, FFmpeg
 - Built an end-to-end pipeline that converts presentation slides into narrated explanatory videos using the FLAN-T5-base transformer model (250 million parameters), deployed on AWS EC2 at an average of 2 minutes per slide (5 minutes per slide with lip-sync avatar mode).
 - Used Amazon Polly for text-to-speech conversion and FFmpeg with Boto3 and Hugging Face APIs to automate end-to-end video rendering and AWS service integration.
- **AgriConnect**
MERN Stack Developer | React.js, Node.js, MongoDB, Gemini API
 - Built a full-stack platform (React.js, Node.js, Express.js, MongoDB) with 5 core modules for farmers, integrating the Government Open Data API (data.gov.in) for real-time, variety-wise market prices and a web crawler for government scheme updates.
 - Used the Google Gemini API to support multi-language translation, making the platform accessible to farmers across multiple Indian languages.
- **AI Powered Disease Prediction Model**
Machine Learning | Flask, REST API
 - Built a machine learning system predicting 5 brain diseases from 5 clinical input parameters, training and comparing 5 classification algorithms (Extra Tree, Naive Bayes, KNN, Decision Tree, Neural Networks) to reach 95% accuracy with the best-performing model.
 - Deployed the trained model as a REST API using Flask, enabling real-time disease prediction from user input.

ACHIEVEMENTS & PARTICIPATIONS

- **Caterpillar CODE-A-THON:** Collaborated in a competitive hackathon to design solutions for real-world engineering problems.
- **DevJams'24 – Google Developer Student Clubs:** Participated in rapid prototyping and collaborative software development.
- **Yantra Central Hack – IEEE Computer Society:** Worked on innovative technical solutions under strict time constraints.
- **CookOff 9.0 – CodeChef:** Competed in a competitive programming contest, improving algorithmic problem-solving skills.